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Dynamics of a Fishery Management Model with Predation-Induced Fear Effect, Impulsive Nonlinear Harvesting Prey and Predator Seasonally Migrating Between Two Patches

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Abstract

This paper proposes a novel fishery management model that integrates three critical ecological factors: the prey (fish) fear effect, impulsive nonlinear harvesting, and predator seasonal migration. It is demonstrated that all solutions of the proposed system are uniformly ultimately bounded. We establish the conditions for the local and global asymptotic stability of the prey-extinction periodic solution, derive the permanence criteria for the system, and determine the threshold condition for prey extinction. Numerical simulations are conducted to validate the theoretical findings. These simulations also help identify the key parameters influencing the threshold condition and reveal the complex dynamics of the system. The results provide significant insights for fishery management, suggesting that regulating harvesting intensity and timing, as well as considering predator migration mortality, are crucial for sustaining fish populations and preventing overexploitation.

Keywords: prey–predator system; fear effect; impulsive density-dependent nonlinear harvesting; large-scale migration; global asymptotic stability; permanence

MSC: 34A37; 34D05; 34D23; 34E05; 37M05

1. Introduction

Fisheries constitute a key economic driver, supported by a complete industrial chain that encompasses the entire process from fishing and aquaculture to processing and sales [1]. Fisheries sustain employment and livelihoods for millions, worldwide. Nevertheless, combined pressures from population growth and rising demand for aquatic products are intensifying the strain on global fishery resources [2,3]. Overfishing and a lack of scientific planning deplete fish stocks, disrupt aquatic food webs, and threaten biodiversity. The urgent implementation of science-based management is thus essential to ensure the long-term sustainability of fishery resources [4,5].

The long-distance migrations undertaken by many birds under climatic and other influences have been comprehensively explored in studies such as [6–8]. Empirical evidence confirms that large-scale bird migration impacts fish resources. Regional studies of Lake Erie (North America), the Dal River (Sweden), Norway, and South Africa demonstrate that migratory birds alter fish population size, distribution, and ecological balance through differential predation pressure [9–11]. Given that these birds predate economically valuable fish species, this behavior can lead to significant population declines [12]. Consequently,

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fishery management models should integrate the impacts of bird migration, necessitating the development of suitable biological models to describe this critical predator–prey dynamic [13].

Biodynamic systems, recognized for their accurate prediction of biological population trends, have attracted considerable worldwide interest [14]. Such systems with impulsive effects are now widely applied across various fields. For instance, Liu et al. [15] examined a Holling-I functional response predator–prey system subject to impulsive linear pesticide spraying and predator release:

$$\begin{cases} \left.\begin{aligned} \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} &= x_1(t)(r - ax_1(t) - bx_2(t)), \\ \frac{\mathrm{d}x_2(t)}{\mathrm{d}t} &= x_2(t)(-d + cx_1(t)), \end{aligned}\right\} & t \neq nT \\ \left.\begin{aligned} \Delta x_1(t) &= -p_1 x_1(t), \\ \Delta x_2(t) &= -p_2 x_2(t) + \mu, \end{aligned}\right\} & t = nT \end{cases}, \tag{1}$$

where $x_1(t)$ stands for the prey (pest) population and $x_2(t)$ for the predator (natural enemy) population. During the non-impulsive phase, the prey population $x_1(t)$ grows at an intrinsic rate $r > 0$. It is restricted by the intraspecific competition coefficient $a > 0$ and the predation rate $b > 0$ of predators. Meanwhile, the predator population $x_2(t)$ has a natural mortality rate $d > 0$ and grows through predation with a conversion rate $c > 0$. Impulsive events occur periodically with period T. At these moments, pesticides are sprayed, instantly reducing the prey population by proportion $0 \leqslant p_1 < 1$ and the predator population by proportion $0 \leqslant p_2 < 1$. $\mu > 0$ predators are artificially added. Ultimately, the global asymptotic stability condition for the prey-extinction periodic solution, the persistence condition of the system, and the existence and stability criteria for the positive periodic solution are derived. In [16], a fishery model with impulsive linear harvesting was explored, establishing the global asymptotic stability conditions for the prey extinction periodic solution and the criteria for system permanence. Jiao et al. [17] investigated an impulsive predator–prey model incorporating seasonal large-scale migration of predators. Other related studies can be found in [18–23]. In terms of nonlinear impulses, Li [24,25] incorporated nonlinear impulsive control into pest management models, and Cheng [26] studied nonlinear impulsive control in a phytoplankton-fish system.

Most models with impulsive effects in fisheries use linear harvesting. In contrast, impulsive density-dependent nonlinear harvesting has received less attention, despite its direct relevance to the population monitoring central to effective management [27–29]. Harvesting efforts should therefore be regulated by monitoring fish population levels. Furthermore, while most existing studies on impulsive predator–prey models rely on Holling-I or Holling-II functional responses that depend solely on prey density, a more ecologically realistic framework requires functional responses influenced by both prey and predator densities [30]. The large-scale migration of piscivorous birds impacts fish populations and fisheries due to their energy dependence on this prey. This interaction, which is bidirectional as fish resource availability indirectly influences bird numbers, is modeled using the Beddington–DeAngelis functional response. This response depends on the densities of both species and quantifies the inhibitory effect of predator interference on predation rates, leading to more accurate predictions that have been validated in several cases [31].

Predator–prey interactions extend beyond direct killing. In reality, prey perceiving threats (fish) may alter their behaviors (e.g., increased concealment), which in turn may affect predators (birds) in terms of search efficiency or handling time. Incorporating such bidirectional behavioral feedback into the model will undoubtedly make the model more

ecologically realistic. However, due to significant challenges in quantifying these trophic-level bidirectional behavioral interactions in modeling and parameter estimation, as a preliminary theoretical exploration, we intentionally focus on the core, well-documented direct effect of fear on the growth rate of the prey population (fish). Long-term fear stress induced by predators can significantly alter prey physiology, life history, and reproduction. This phenomenon is now central to ecological research [32]. This is demonstrated by diverse cases: wolves altering elk metabolism [33], odors of mantises causing harm to fruit flies [34], and predator cues causing PTSD-like (post-traumatic stress disorder) impairments that compromise reproduction in chickadees [35]. Beyond traditional predators, the large-scale migration of birds constitutes a significant source of fear stress for aquatic prey like fish [36]. This stress elicits a physiological response, elevating cortisol and other stress hormones via the HPA axis (hypothalamic-pituitary-adrenal axis), which consequently impairs immune function, disrupts reproductive hormones, and suppresses reproduction [37,38]. Prolonged exposure can also alter metabolic scope, hinder growth, and ultimately lead to reduced birth rates [39]. Evidence shows that the predation threat from migratory birds can disrupt fish metabolism and behavior, impair reproduction through energetic and hormonal pathways, and ultimately reduce reproductive success [40–42]. The fear effect on fish is still overlooked in fishery models, unlike other ecological factors.

Based on this, this paper establishes a novel Beddington–DeAngelis functional response fishery management system with fear effect, predator large-scale migration and impulsive density-dependent nonlinear harvesting prey (fish),

$$\begin{cases} \left.\begin{aligned} \frac{\mathrm{d}x_0(t)}{\mathrm{d}t} &= x_0(t)(a_{10} - b_{10}x_0(t)), \\ \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} &= x_1(t)(a_{11} - b_{11}x_1(t)), \end{aligned}\right\} t \in (nP, (n+\theta)P] \\ \left.\begin{aligned} \Delta x_0(t) &= -\frac{h_{max}}{\alpha_0 + x_0(t)}x_0(t), \\ \Delta x_1(t) &= -d_0 x_1(t), \end{aligned}\right\} t = (n+\theta)P \\ \left.\begin{aligned} \frac{\mathrm{d}x_0(t)}{\mathrm{d}t} &= x_0(t)\left(\frac{a_{20}}{1+\alpha x_1(t)} - b_{20}x_0(t)\right) - \frac{\zeta_0 x_0(t)x_1(t)}{1 + A_0 x_0(t) + B_0 x_1(t)}, \\ \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} &= -dx_1(t) + \frac{c_0\zeta_0 x_0(t)x_1(t)}{1 + A_0 x_0(t) + B_0 x_1(t)}, \end{aligned}\right\} t \in ((n+\theta)P, (n+1)P] \\ \left.\begin{aligned} \Delta x_0(t) &= 0, \\ \Delta x_1(t) &= -d_1 x_1(t), \end{aligned}\right\} t = (n+1)P \end{cases} . \tag{2}$$

A seasonal migration pattern is assumed for predators moving between patch 1 and patch 2. The density $x_0(t)$ describes the prey population, which is confined to patch 2. The predator population density is represented by $x_1(t)$. For prey population $x_0(t)$ in patch 2 over $(nP, (n+\theta)P]$, $a_{10} > 0$ corresponds to the inherent growth rate; similarly, $b_{10} > 0$ represents the intraspecific competition coefficient. $\frac{h_{\max}}{\alpha_0 + x_0(t)}x_0(t)$ denotes the impulsive density-dependent nonlinear harvesting of prey (fish) at time $t = (n+\theta)P$. The coefficient $h_{\max}$, where $h_{\max} > 0$, represents the maximum harvest amount. That is, prey harvesting happens right before predators move in large numbers seasonally from patch 1 to patch 2. This happens at time $t = (n+\theta)P$. Furthermore, the maximum amount of prey (fish) is harvested when the prey quantity $x_0(t)$ approaches infinity. On the contrary, no harvest is made when the prey quantity approaches 0. The mortality coefficient d_0 $(0 < d_0 < 1)$ governs the population loss that predators $x_1(t)$ undergo during their seasonal migration from patch 1 to patch 2 at time $t = (n+\theta)P$. In patch 2 over $((n+\theta)P, (n+1)P]$, $a_{20} > 0$ denotes the inherent growth rate of prey $x_0(t)$, while $b_{20} > 0$ denotes its intraspecific competition coefficient in the same period. The fear effect is characterized by the birth

rate term $\frac{a_{20}}{1+\alpha x_1(t)}$, with $\alpha > 0$ quantifying the fear level. Consequently, an increase in α leads to a reduction in the prey birth rate. The term $\frac{\zeta_0 x_0(t)x_1(t)}{1+A_0x_0(t)+B_0x_1(t)}$ characterizes the Beddington–DeAngelis functional response, capturing how predator capture ability depends on both prey(fish) and predator densities, rather than prey density alone. Over the interval $((n+\theta)P, (n+1)P]$, $\zeta_0 > 0$ quantifies the prey capture ability by predators in patch 2, and $c_0 > 0$ denotes the associated conversion rate into predator growth. $A_0 > 0$ represents the handling time on the feeding rate and $B_0 > 0$ characterizes the intensity of interference among predators. The coefficient d_1 $(0 < d_1 < 1)$ represents the mortality rate during the seasonal large-scale migration of predators $x_1(t)$ from patch 2 to patch 1 at $t = (n+1)P$. These migratory events occur periodically every P time units, where $P > 0$ is a constant and $0 < \theta < 1$. Additionally, the differences $\Delta x_0(t)$ and $\Delta x_1(t)$ are defined as $\Delta x(t) = x_0(t^+) - x_0(t)$ and $\Delta x_1(t) = x_1(t^+) - x_1(t)$, respectively.

2. Preliminaries

Remark 1. *In system (2), the dynamics of the predator population $x_1(t)$ in patch 2 are affected by its seasonal immigration and emigration (characterized by impulses) as well as local mortality. The prey-extinction periodic solution $(0, \widetilde{x}_1(t))$ focused on in this paper describes the state of the predator population after the collapse of the fish resource $x_0(t)$ in patch 2. Mathematically, this solution arises from the subsystem (3) when $x_0(t) = 0$, where predator dynamics are driven solely by intrinsic growth during the migration-free period $(nP, (n+\theta)P]$, natural mortality during the predation period $((n+\theta)P, (n+1)P]$, and impulsive losses d_0, d_1 during migration events. The condition $(1-d_0)(1-d_1)e^{a_{11}\theta P - d(1-\theta)P} > 1$ ensures a positive periodic solution $\tilde{x}_1(t)$, which is sustained not by local prey but by the balance between growth in patch 1 and mortality during migration. This mathematical construct is used to model the ecological scenario where predators are sustained by external resources (e.g., other prey in patch 1) and continue to exert periodic predation pressure on patch 2 even after local prey collapse. From a biological perspective, the sustained oscillations of $x_1(t)$ in this solution do not imply that predators can persist independently of local fish resources for a long time, but rather reflect the following reality: after the depletion of the target fishery population (prey), the predation pressure from migratory populations outside the system will still periodically occur. In fishery management models, the core purpose of analyzing the stability of this solution is to accurately define the critical threshold at which the target fish population moves towards commercial extinction (unable to support sustainable fishing), rather than predicting the long-term independent survival of the predator population.*

Consider system (2), whose right-hand side defines a vector mapping $p = (p_1, p_2)$. Let $\mathbb{R}_+ = [0, \infty)$ and $\mathbb{R}_+^2 = \{(m_1, m_2) \mid m_1 > 0, m_2 > 0\}$. A solution to this system can be expressed as $F(t) = (x_0(t), x_1(t))^P$, where the domain is $\mathbb{R}_+$ and the range belongs to $\mathbb{R}_+^2$. This solution is continuous on the intervals $(nP, (n+\theta)P]$ and $((n+\theta)P, (n+1)P]$ $(n \in \mathbb{Z}^+)$, and possesses right limits at pulse instants: $x_0((n+\theta)P^+) = lim_{t\to(n+\theta)P^+} x_0(t)$, $x_1((n+\theta)P^+) = lim_{t\to(n+\theta)P^+} x_1(t)$, $x_0((n+1)P^+) = lim_{t\to(n+1)P^+} x_0(t)$, $x_1((n+1)P^+) = lim_{t\to(n+1)P^+} x_1(t)$. Thus, $F(t)$ is a piecewise continuous function. According to the description in [43], the p is smooth, ensuring global uniqueness of solutions to the system (2). We state several key lemmas that form the basis for our subsequent theoretical developments.

Lemma 1. *If the solution $F(t)$ of system (2) satisfies $F(0^+) \geqslant 0$, then $F(t) \geqslant 0$ for all $t \geqslant 0$. Additionally, if $F(0^+) > 0$, then $F(t) > 0$ for all $t \geqslant 0$.*

Lemma 2. *There exists a constant $J_1 > 0$ such that $x_0(t) \leqslant J_1$ and $x_1(t) \leqslant J_1$ for all sufficiently large t, where $(x_0(t), x_1(t))$ denotes a solution of system (2).*

Proof. Suppose $V(t) = c_0x_0(t) + x_1(t)$, we will obtain $t \in (nP, (n+\theta)P]$. By calculating the upper right Dini derivative of $V(t)$, we obtain

$$\begin{aligned} D^+V(t) + kV(t) &= c_0(a_{10}+d)x_0(t) - c_0b_{10}x_0^2(t) + (a_{11}+d)x_1(t) - b_{11}x_1^2(t) \\ &= -c_0b_{10}\left(x_0(t) - \frac{a_{10}+k}{2b_{10}}\right)^2 + \frac{c_0(a_{10}+k)^2}{4b_{10}} \\ &- b_{11}\left(x_1(t) - \frac{a_{11}+k}{2b_{11}}\right)^2 + \frac{(a_{11}+k)^2}{4b_{11}} \\ &\leqslant \frac{c_0(a_{10}+k)^2}{4b_{10}} + \frac{(a_{11}+k)^2}{4b_{11}}, \end{aligned}$$

when $t = (n+\theta)P$,

$$V(t^+) = c_0x_0(t)\left(1 - \frac{h_{\max}}{\alpha_0 + x_0(t)}\right) + (1-d_0)x_1(t) \leqslant V(t),$$

when $t \in ((n+\theta)P, (n+1)P]$,

$$\begin{aligned} D^+V(t) + kV(t) &= -c_0b_{20}x_0^2(t) + c_0x_0(t)\left(\frac{a_{20}}{1+\alpha x_1(t)} + k\right) + (k-d)x_1(t) \\ &\leqslant -c_0b_{20}x_0^2(t) + c_0x_0(t)(a_{20}+k) \\ &= -c_0b_{20}\left(x_0(t) - \frac{a_{20}+k}{2b_{20}}\right)^2 + \frac{c_0(a_{20}+k)^2}{4b_{20}}, \end{aligned}$$

when $t = (n+1)P$,

$$V(t^+) = c_0x_0(t) + (1-d_1)x_1(t) \leqslant V(t).$$

Letting $M = min\left\{\frac{c_0(a_{10}+k)^2}{4b_{10}}, \frac{(a_{11}+k)^2}{4b_{11}}, \frac{c_0(a_{20}+k)^2}{4b_{20}}\right\}$, we derive

$$\begin{cases} V(t) + kV(t) \leqslant M & ,t \in (nP, (n+1)P] \\ V(t^+) \leqslant V(t) & ,t = (n+1)P \end{cases}.$$

By applying the comparison theorem for impulsive differential equations, we derive

$$V(t) \leqslant \left(V(0^+) - \frac{M}{k}\right)exp(-nkP) + \frac{M}{k} \to \frac{M}{k}(t \to \infty).$$

Consequently, it can be deduced that $V(t)$ is uniformly ultimately bounded. Furthermore, by the definition of $V(t)$, $x_0(t)$ and $x_1(t)$ possess an upper bound J_1 such that $x_0(t) \leqslant J_1$ and $x_1(t) \leqslant J_1$. □

Letting $x_0(t) = 0$, the subsystem of system (2) can be obtained as follows:

$$\begin{cases} \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = x_1(t)(a_{11} - b_{11}x_1(t)) & , t \in (nP, (n+\theta)P] \\ \Delta x_1(t) = -d_0x_1(t) & , t = (n+\theta)P \\ \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = -dx_1(t) & , t \in ((n+\theta)P, (n+1)P] \\ \Delta x_1(t) = -d_1x_1(t) & , t = (n+1)P \end{cases}. \tag{3}$$

The analytical solution follows

$$x_1(t) = \begin{cases} \frac{a_{11}x_1(nP^+)e^{a_{11}(t-nP)}}{a_{11}+b_{11}x_1(nP^+)(e^{a_{11}(t-nP)}-1)} & ,t \in (nP,(n+\theta)P] \\ x_1((n+\theta)P^+)e^{-d(t-(n+\theta)P)} & ,t \in ((n+\theta)P,(n+1)P] \end{cases}. \tag{4}$$

Equation (4) has a trivial fixed point $x^* = 0$. Moreover, under the condition $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} > 1$, it also admits a positive fixed point $x^* = \frac{a_{11}((1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}-1)}{b_{11}e^{a_{11}\theta P}-1)}$. The periodic solution of system (3) is

$$\widetilde{x}_1(t) = \begin{cases} \frac{a_{11}x^*e^{a_{11}(t-nP)}}{a_{11}+b_{11}x^*(e^{a_{11}(t-nP)}-1)} & ,t \in (nP,(n+\theta)P] \\ x^{**}e^{-d(t-(n+\theta)P)} & ,t \in ((n+\theta)P,(n+1)P] \end{cases}.$$

where

$$\begin{cases} x^* = \frac{a_{11}((1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}-1)}{b_{11}(e^{a_{11}\theta P}-1)} \\ x^{**} = \frac{(1-d_0)a_{11}x^*e^{a_{11}\theta P}}{a_{11}+b_{11}x^*(e^{a_{11}\theta P}-1)} \end{cases}.$$

Lemma 3. *(i) The trivial fixed point is globally asymptotically stable when*

$$(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} < 1.$$

(ii) The positive fixed point x^* is globally asymptotically stable when

$$(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} > 1.$$

Proof. Letting $x_n = x_1(nP^+)$, the stroboscopic map of system (2) is

$$\begin{aligned} x_{n+1} &= \frac{(1-d_0)(1-d_1)a_{11}e^{a_{11}\theta P-d(1-\theta)P}x_n}{a_{11}+b_{11}(e^{a_{11}\theta P}-1)x_n} \\ &= f(x_n), \\ \frac{\mathrm{d}f(x_n)}{\mathrm{d}x_n} &= \frac{a_{11}^2(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}}{(a_{11}+b_{11}(e^{a_{11}\theta P}-1)x_n)^2}. \end{aligned}$$

When $x^* = 0$,

$$\left|\left.\frac{\mathrm{d}f(x_n)}{\mathrm{d}x_n}\right|_{x_n=x^*}\right| = (1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} < 1,$$

if $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} < 1$. When $x^* \neq 0$,

$$\left|\left.\frac{\mathrm{d}f(x_n)}{\mathrm{d}x_n}\right|_{x_n=x^*}\right| = \frac{1}{(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}} < 1,$$

if $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} > 1$. Next, we will prove the global stability. We have

$$|x_{n+1}-0| = \frac{a_{11}(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}x_n}{a_{11}+b_{11}(e^{a_{11}\theta P}-1)x_n} \leqslant (1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}x_n.$$

That is, the trivial fixed point is globally asymptotically stable if $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} < 1$.

For the positive fixed point, we have

$$\left|\frac{1}{x_{n+1}}-\frac{1}{x^*}\right| = \left|\frac{a_{11}+b_{11}\left(e^{a_{11}\theta P}-1\right)x_n}{(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}x_n} - \frac{a_{11}+b_{11}\left(e^{a_{11}\theta P}-1\right)x^*}{(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}x^*}\right|$$
$$= \frac{1}{(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P}}\left|\frac{1}{x_n}-\frac{1}{x^*}\right| \to 0 \text{ as } n\to\infty.$$

Therefore, the positive fixed point x^* is globally asymptotically stable if $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} > 1$. □

Similar to Ref. [25], we can obtain the following lemma:

Lemma 4. *The periodic solution $\widetilde{x}_1(t)$ becomes globally asymptotically stable when the condition*

$$(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} > 1$$

is satisfied.

Consider the following conditions

(H_1) $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P} > 1$,

(H_2) $\left(1-\frac{h_{\max}}{\alpha_0}\right)\exp(a_{10}\theta P+d(1-\theta)P+\mathbf{M}) < 1$,

(H_2') $(1-d_0)(1-d_1)e^{a_{11}\theta P-d(1-\theta)P-2\ln\frac{a_{11}+b_{11}x^*(e^{a_{11}\theta P}-1)}{a_{11}}} < 1$,

(H_3) $\left(1-\frac{h_{\max}}{\alpha_0+b}\right)\exp(a_{10}\theta P+d(1-\theta)P+\mathbf{M}') < 1$,

(H_4) $\left(1-\frac{h_{\max}}{\alpha_0}\right)\exp(a_{10}\theta P+d(1-\theta)P+\mathbf{M}) > 1$,

with

$$\mathbf{M} = \ln\frac{1+\alpha x^{**}e^{-d(1-\theta)P}}{1+\alpha x^{**}} + \frac{\zeta_0}{dB_0}\ln\frac{1+B_0x^{**}e^{-d(1-\theta)P}}{1+B_0x^{**}},$$
$$\mathbf{M}' = \ln\frac{1+\alpha x^{**}e^{-d(1-\theta)P}}{1+\alpha x^{**}} + \frac{\zeta_0}{dB_0}\ln\frac{1+A_0b+B_0x^{**}e^{-d(1-\theta)P}}{1+A_0b+B_0x^{**}}.$$

3. Main Results

In bio-dynamical systems, investigating the stability of the prey-extinction periodic solution is not aimed at achieving such a state, but rather to pinpoint the critical threshold at which the system collapses, thereby providing an early warning for irreversible ecological or economic consequences. This section focuses on analyzing the stability of the prey-extinction periodic solution $(0, \widetilde{x}_1(t))$ of system (2): if this solution is stable, it implies that once the prey population density falls below a certain critical level, the system will irreversibly decline toward prey extinction. Notably, the periodic dynamics of the predator population in this solution characterize the ecological reality that external predation pressure still persists periodically even after the depletion of the target fish resource. This is mathematically represented by the predator subsystem (3), which is decoupled from prey dynamics and sustained by migration-driven growth and mortality balance, rather than local prey consumption. Consequently, the stability condition of this solution corresponds to a core threshold for preventing the collapse of the target fish population in management practice, offering crucial early warning support for fishery resource risk management.

3.1. Stability

To mathematically define the extinction threshold for the prey population, we analyze the stability of the boundary periodic solution $(0, \widetilde{x}_1(t))$. The stability of this solution implies that once the prey density falls below a critical level due to overharvesting or predation pressure, the system will irreversibly converge to a state where the prey is extinct—a scenario that must be avoided in management.

Theorem 1. *If conditions (H_1), (H_2), and (H_2') hold, then the periodic solution $(0, \widetilde{x}_1(t))$ of system (2) is locally asymptotically stable.*

Proof. Define $u(t) = x_0(t)$, $v(t) = x_1(t) - \widetilde{x}_1(t)$, the following linear approximation system is obtained for system (2)

$$\begin{pmatrix} \frac{\mathrm{d}u(t)}{\mathrm{d}t} \\ \frac{\mathrm{d}v(t)}{\mathrm{d}t} \end{pmatrix} = \begin{pmatrix} a_{10} & 0 \\ 0 & a_{11} - 2b_{11}\widetilde{x}_1(t) \end{pmatrix} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}, t \in (nP, (n+\theta)P],$$

and

$$\begin{pmatrix} \frac{\mathrm{d}u(t)}{\mathrm{d}t} \\ \frac{\mathrm{d}v(t)}{\mathrm{d}t} \end{pmatrix} = \begin{pmatrix} \frac{a_{20}}{1+\alpha\widetilde{x}_1(t)} - \frac{\zeta_0\widetilde{x}_1(t)}{1+B_0\widetilde{x}_1(t)} & 0 \\ \frac{c_0\zeta_0\widetilde{x}_1(t)}{1+B_0\widetilde{x}_1(t)} & -d \end{pmatrix} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}, t \in ((n+\theta)P, (n+1)P].$$

For $t \in (nP, (n+\theta)P]$, the fundamental solution matrix can be readily calculated

$$Y_1(t) = \begin{pmatrix} \exp(a_{10}(t - nP)) & 0 \\ * & \exp\left(\int_{nP}^{t}(a_{11} - 2b_{11}\widetilde{x}_1(s))\mathrm{d}s\right) \end{pmatrix}.$$

For $t \in ((n+\theta)P, (n+1)P]$, the fundamental solution matrix is as follows

$$Y_2(t) = \begin{pmatrix} \exp\left(\int_{(n+\theta)P}^{t}\left(\frac{a_{20}}{1+\alpha\widetilde{x}_1(s)} - \frac{\zeta_0\widetilde{x}_1(s)}{1+B_0\widetilde{x}_1(s)}\right)\mathrm{d}s\right) & 0 \\ * & \exp(-d(t - (n+\theta)P)) \end{pmatrix}.$$

Since the specific expression denoted by $*$ is not used in the subsequent analysis, it is omitted. The linearization of the third and fourth equations in system (2) is given by

$$\begin{pmatrix} u((n+\theta)P^+) \\ v((n+\theta)P^+) \end{pmatrix} = \begin{pmatrix} 1 - \frac{h_{\max}}{\alpha_0} & 0 \\ 0 & 1 - d_0 \end{pmatrix} \begin{pmatrix} u((n+\theta)P) \\ v((n+\theta)P) \end{pmatrix},$$

and the linearization of the seventh and eighth equations in system (2) is given by

$$\begin{pmatrix} u((n+1)P^+) \\ v((n+1)P^+) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 - d_1 \end{pmatrix} \begin{pmatrix} u((n+1)P) \\ v((n+1)P) \end{pmatrix}.$$

The local stability of the prey-extinction periodic solution $(0, \widetilde{x}_1(t))$ can be determined by the eigenvalues of the matrix

$$
\begin{aligned}
G &= \begin{pmatrix} 1 & 0 \\ 0 & 1-d_1 \end{pmatrix} Y_2(P) \begin{pmatrix} 1-\frac{h_{max}}{\alpha_0} & 0 \\ 0 & 1-d_0 \end{pmatrix} Y_1(\theta P) \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1-d_1 \end{pmatrix} \begin{pmatrix} \exp\left(\int_{\theta P}^{P}\left(\frac{a_{20}}{1+\alpha\widetilde{x}_1(s)} - \frac{\zeta_0\widetilde{x}_1(s)}{1+B_0\widetilde{x}_1(s)}\right)ds\right) & 0 \\ * & \exp(-d(1-\theta)P) \end{pmatrix} \\
&\times \begin{pmatrix} 1-\frac{h_{max}}{\alpha_0} & 0 \\ 0 & 1-d_0 \end{pmatrix} \begin{pmatrix} \exp(a_{10}\theta P) & 0 \\ 0 & \exp\left(\int_0^{\theta P}(a_{11}-2b_{11}\widetilde{x}_1(s))\mathrm{d}s\right) \end{pmatrix} \\
&= \begin{pmatrix} \left(1-\frac{h_{max}}{\alpha_0}\right)\exp\left(\int_{\theta P}^{P}\left(\frac{a_{20}}{1+\alpha\widetilde{x}_1(s)} - \frac{\zeta_0\widetilde{x}_1(s)}{1+B_0\widetilde{x}_1(s)}\right)\mathrm{d}s + a_{10}\theta P\right) & 0 \\ \divideontimes & G_{22} \end{pmatrix},
\end{aligned} \tag{5}
$$

where $G_{22} = (1-d_0)(1-d_1)\exp\left(\int_0^{\theta P}(a_{11}-2b_{11}\widetilde{x}_1(s))\mathrm{d}s - d(1-\theta)P\right)$.

It is straightforward to determine that the matrix G possesses two eigenvalues

$$
\lambda_1 = \left(1-\frac{h_{\max}}{\alpha_0}\right)\exp\left(\int_{\theta P}^{P}\left(\frac{a_{20}}{1+\alpha\widetilde{x}_1(s)} - \frac{\zeta_0\widetilde{x}_1(s)}{1+B_0\widetilde{x}_1(s)}\right)\mathrm{d}s + a_{10}\theta P\right),
$$

and

$$
\lambda_2 = (1-d_0)(1-d_1)\exp\left(\int_0^{\theta P}(a_{11}-2b_{11}\widetilde{x}_1(s))\mathrm{d}s - d(1-\theta)P\right).
$$

When the condition specified in Theorem 1 is satisfied, we can derive $|\lambda_1| < 1$ and $|\lambda_2| < 1$. By virtue of the Floquet theorem [43], the boundary periodic solution $(0, \widetilde{x}_1(t))$ is locally asymptotically stable. □

Denote

$$
\rho_0 = a_{10}\theta P + \int_{(n+\theta)P}^{(n+1+\theta)P}\left(\frac{a_{20}}{1+\alpha(\widetilde{x}_1(s)-\varepsilon_1)} - \frac{\zeta_0(\widetilde{x}_1(s)-\varepsilon_1)}{1+A_0(b+\epsilon)+B_0(\widetilde{x}_1(s)-\varepsilon_1)}\right)\mathrm{d}s.
$$

Theorem 2. *The global asymptotic stability of $(0, \widetilde{x}_1(t))$ is guaranteed when conditions (H_1), (H_2') and (H_3) hold simultaneously.*

Proof. We demonstrate the global attraction of the solution. Lemma 4 implies that for any $\varepsilon_1 > 0$, there exists $n_0P > 0$ satisfying

$$
x_1(t) \geqslant \widetilde{x}_1(t) - \varepsilon_1, \quad \forall t > n_0P.
$$

Define $b = \max\left\{\frac{a_{10}}{b_{10}}, \frac{a_{20}}{b_{20}}\right\}$. We choose $\varepsilon_1 > 0$ and $d > 0$ to ensure $\rho < 1$ with $\rho = e^{\rho_0}\left(1-\frac{h_{\max}}{\alpha_0+b+\epsilon_1}\right)$. Now, considering system (2), the following inequality holds

$$
\frac{\mathrm{d}x_0}{\mathrm{d}t} \leqslant x_0\left(a_{10} - b_{10}x_0 - \frac{h_{\max}}{\alpha_0+b+\varepsilon_1}\right).
$$

Subsequently,

$$
\begin{cases}
\frac{\mathrm{d}x_0(t)}{\mathrm{d}t} \leqslant a_{10}x_0(t) & ,t \in (nP, (n+\theta)P] \\
\Delta x_0(t) \leqslant -\frac{h_{\max}}{\alpha_0+b+\epsilon}x_0(t) & ,t = (n+\theta)P \\
\frac{\mathrm{d}x_0(t)}{\mathrm{d}t} \leqslant \frac{a_{20}x_0(t)}{1+\alpha(\widetilde{x}_1(t)-\varepsilon_1)} - \frac{\zeta_0(\widetilde{x}_1(t)-\varepsilon_1)x_0(t)}{1+A_0(b+\epsilon)+B_0(\widetilde{x}_1(t)-\varepsilon_1)} & ,t \in ((n+\theta)P, (n+1)P] \\
\Delta x_0(t) = 0 & ,t = (n+1)P
\end{cases}.
$$

Thus, we obtain the inequality

$$x_0((n+1+\theta)P^+) \leqslant \rho x_0((n+\theta)P^+).$$

This implies that for any integer $k_0 > 0$,

$$x_0((n_1+k_0+\theta)P) \leqslant e^{k_0\rho} x_0((n_1+\theta)P^+).$$

As $k_0 \to +\infty$, the right-hand side tends to 0, thus,

$$x_0(t) \to 0 \quad \text{as} \quad t \to \infty.$$

We now prove that $x_1(t) \to \widetilde{x}_1(t)$ as $t \to +\infty$. For any $\varepsilon > 0$ with $0 < \varepsilon < \dfrac{\alpha}{c_0\zeta_0 d A_0}$, there exists $n_1 P > 0$ such that for all $t \geqslant n_1 P$, we have $x_0(t) \leqslant \varepsilon$.

From system (2), we derive the following equations:

$$\begin{cases} \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = x_1(t)(a_{11} - b_{11}x_1(t)) & ,t \in (nP,(n+\theta)P] \\ \Delta x_1(t) = -d_0 x_1(t) & ,t = (n+\theta)P \\ \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} \leqslant \left(-d + \frac{c_0\zeta_0\varepsilon}{1+A_0\varepsilon}\right)x_1(t) & ,t \in ((n+\theta)P,(n+1)P] \\ \Delta x_1(t) = -d_1 x_1(t) & ,t = (n+1)P \end{cases}.$$

Applying the comparison theorem of impulsive differential equations [43], we arrive at the following system of equations:

$$\begin{cases} \frac{\mathrm{d}x_3(t)}{\mathrm{d}t} = x_3(t)(a_{11} - b_{11}x_3(t)) & ,t \in (nP,(n+\theta)P] \\ \Delta x_3(t) = -d_0 x_3(t), & ,t = (n+\theta)P \\ \frac{\mathrm{d}x_3(t)}{\mathrm{d}t} = \left(-d + \frac{c_0\zeta_0\varepsilon}{1+A_0\varepsilon}\right)x_3(t) & ,t \in ((n+\theta)P,(n+1)P] \\ \Delta x_3(t) = -d_1 x_3(t) & ,t = (n+1)P \end{cases}.$$

Consequently, the periodic solution $\widetilde{x}_3(t)$ of the above system is derived as follows:

$$\widetilde{x}_3(t) = \begin{cases} \frac{a_{11}x_3^* e^{a_{11}(t-nP)}}{a_{11}+b_{11}x_3^*(e^{a_{11}(t-nP)}-1)} & ,t \in (nP,(n+\theta)P] \\ x_3^{**} e^{(-dt+\frac{c_0\zeta_0\varepsilon}{1+A_0\varepsilon})(t-(n+\theta)P)} & ,t \in ((n+\theta)P,(n+1)P] \end{cases}.$$

where

$$\begin{cases} x_3^* = \frac{\mathbf{Q}a_{11}}{b_{11}(e^{a_{11}\theta P}-1)} \\ x_3^{**} = (1-d_0)\frac{a_{11}x_3^* e^{a_{11}\theta P}}{a_{11}+b_{11}x_4^*(e^{a_{11}\theta P}-1)} \end{cases}.$$

with $\mathbf{Q} = (1-d_0)(1-d_1)e^{a_{11}\theta P + \left(-d+\frac{c_0\zeta_0\varepsilon}{1+A_0\varepsilon}\right)(1-\theta)P}$.

Based on the global asymptotic stability of the periodic solution $\widetilde{x}_1(t)$ established in Lemma 4, for any given $\varepsilon_1 > 0$, there exists $n_0 P > 0$ such that when $t > n_0 P$, the inequality $x_1(t) \geqslant \widetilde{x}_1(t) - \varepsilon_1$ holds. For any $\varepsilon_2 > 0$, there exists t_1 such that when $t > t_1$,

$$\widetilde{x}_1(t) - \varepsilon_1 \leqslant x_1(t) \leqslant \widetilde{x}_3(t) + \varepsilon_2.$$

Let $\varepsilon \to 0$. Then, for large t,

$$\widetilde{x}_1(t) - \varepsilon_1 < x_1(t) < \widetilde{x}_1(t) + \varepsilon_1,$$

so $x_1(t) \to \widetilde{x}_1(t)$ as $t \to +\infty$. The proof is complete. □

Proving the permanence of the system, on the other hand, ensures that all populations can be maintained within a positive density range over the long term, enabling sustainable coexistence. This corresponds to the ideal objectives of ecosystem health and sustainable resource utilization. Together, these two aspects delineate both the safety boundaries and sustainable pathways for ecological management.

3.2. Permanence

Complementary to the extinction analysis, it is crucial to identify the positive conditions for sustainable management. Permanence ensures that all population densities remain positive in the long term, corresponding to the desired state of sustainable resource utilization. This subsection aims to establish the mathematical criteria for system permanence, thereby defining the safe operating space for management interventions.

Theorem 3. *Under conditions (H_1) and (H_4), system (2) exhibits permanence.*

Proof. An application of Lemma 4 yields

$$x_1(t) \geqslant \widetilde{x}_1(t) - \varepsilon_1 \geqslant x^{**} e^{-d(1-\theta)P} - \varepsilon_1 = \omega_2.$$

Select $\omega_3 > 0$ such that $\omega_3 < \min\left\{\frac{a_{10}}{b_{10}}, \frac{d}{c_0\zeta_0 - A_0 d}\right\}$ and $\rho_2 > 1$, where

$$\rho_2 = (1 - \frac{h_{\max}}{\alpha_0}) e^{(a_{10} - b_{10}\omega_3)lP + \int_{(n+\theta)P}^{(n+1+\theta)P} \left(\frac{a_{20}}{1+\alpha_0(\widetilde{x}_4(s)+\varepsilon)} - b_{20}\omega_3 - \frac{\zeta_0(\widetilde{x}_4(s)+\varepsilon)}{1+B_0(\widetilde{x}_4(s)+\varepsilon)}\right) \mathrm{d}s}.$$

We now establish a uniform positive lower bound for the prey population: there exists $\omega_1 > 0$ such that $x_0(t) \geqslant \omega_1$ holds for all large t.

Moving forward, our task is to prove that there exist $t_1 > 0$, ensuring that $x_0(t_1) \geqslant \omega_3$. Otherwise, for all $t > 0$, we have $x_0(t) < \omega_3$, based on system (2), we can gain

$$\begin{cases} \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} = x_1(t)(a_{11} - b_{11}x_1(t)) & ,t \in (nP, (n+\theta)P) \\ \Delta x_1(t) = -d_0 x_1(t) & ,t = (n+\theta)P \\ \frac{\mathrm{d}x_1(t)}{\mathrm{d}t} \leqslant \left(-d + \frac{c_0\zeta_0\omega_3}{1+A_0\omega_3}\right) x_1(t) & ,t \in ((n+\theta)P, (n+1)P) \\ \Delta x_1(t) = -d_1 x_1(t) & ,t = (n+1)P \end{cases}.$$

Considering comparative system with $x_4(0^+) = x_1(0^+)$,

$$\begin{cases} \frac{\mathrm{d}x_4(t)}{\mathrm{d}t} = x_4(t)(a_{11} - b_{11}x_4(t)) & ,t \in (nP, (n+\theta)P) \\ \Delta x_4(t) = -d_0 x_4(t) & ,t = (n+\theta)P \\ \frac{\mathrm{d}x_4(t)}{\mathrm{d}t} = \left(-d + \frac{c_0\zeta_0\omega_3}{1+A_0\omega_3}\right) x_4(t) & ,t \in ((n+\theta)P, (n+1)P) \\ \Delta x_4(t) = -d_1 x_4(t) & ,t = (n+1)P \end{cases}. \tag{6}$$

Therefore, the periodic solution $\widetilde{x}_4(t)$ of system (6) is derived as follows:

$$\widetilde{x}_4(t) = \begin{cases} \frac{a_{11}x_4^* e^{a_{11}(t-nP)}}{a_{11} + b_{11}x_4^*(e^{a_{11}(t-nP)} - 1)} & ,t \in (nP, (n+\theta)P) \\ x_4^{**} e^{\left(-d + \frac{c_0\zeta_0\omega_3}{1+A_0\omega_3}\right)(t-(n+\theta)P)} & ,t \in ((n+\theta)P, (n+1)P) \end{cases}.$$

where

$$\begin{cases} x_4^* = \frac{\mathbf{P}a_{11}}{b_{11}(e^{a_{11}\theta P}-1)} \\ x_4^{**} = (1-d_0)\frac{a_{11}x_4^* e^{a_{11}\theta P}}{a_{11}-b_{11}x_4^*(e^{a_{11}\theta P}-1)} \end{cases},$$

with $\mathbf{P} = (1-d_0)(1-d_1)e^{a_{11}\theta P+\left(-d+\frac{c_0\zeta_0\omega_3}{1+A_0\omega_3}\right)(1-\theta)P}$. The periodic solution $\widetilde{x}_4(t)$ is globally asymptotically stable from Lemma 4. Therefore, according to comparison theorem of impulsive differential equations [43], for any $\varepsilon > 0$, there exists a t_0 such that for $t > t_0$, $x_1(t) \leqslant x_4(t) \leqslant \widetilde{x}_4(t) + \varepsilon$. From system (2) again, we get

$$\begin{cases} \frac{\mathrm{d}x_0}{\mathrm{d}t} \geqslant x_0(t)(a_{10} - b_{10}\omega_3) & ,t \in (nP, (n+\theta)P) \\ \Delta x_0 \geqslant -\frac{h_{\max}}{\alpha_0}x_0 & ,t = (n+\theta)P \\ \frac{\mathrm{d}x_0}{\mathrm{d}t} \geqslant x_0\left(\frac{a_{20}}{1+\alpha(\widetilde{x}_4(t)+\varepsilon)} - b_{20}\omega_3 - \frac{\zeta_0(\widetilde{x}_4(t)+\varepsilon)}{1+B_0(\widetilde{x}_4(t)+\varepsilon)}\right) & ,t \in ((n+\theta)P, (n+1)P) \end{cases}. \tag{7}$$

Then $x_0((n+1+P)P^+) \geqslant \rho_2 x_0((n+P)P^+)$, for all $t > n_1P$, where $n_1P \geqslant t_0$. Therefore, $x_0((n_1+k_1+\theta)P) \geqslant k_1\rho_2^{n_1}x_0((n_1+\theta)P^+) \to +\infty$ as $k_1 \to +\infty$, which implies $x_0(t) \to +\infty$ as $t \to +\infty$, which is contradiction. Thus there exists a $t_1 > 0$ such that $x_0(t_1) \geqslant \omega_3$.

If $x_0(t) \geqslant \omega_3$ for all $t \geqslant t_1$, then our objective is achieved. We therefore consider only those solutions whose trajectories leave and later re-enter the region defined by $R = \{x \in R_+^2 : x_0 < \omega_3\}$. Let $t^* = \inf_{t \geqslant t_1}\{x_0(t) < \omega_3\}$. Then $x_0(t) \geqslant \omega_3$ for $t \in [t_1, t^*)$.

Case A: with $t^* = (n_2+\theta)P$. Select natural numbers $n_3, n_4 \in \mathbb{N}$ such that

$$\rho_3 = \left(1 - \frac{h_{\max}}{\alpha_0}\right)^{n_3} e^{n_3\delta_3 P}\rho_2^{n_4} > 1,$$

with the conditions $\delta_1 = a_{10} - b_{10}\omega_3 < 0$, $\delta_2 = \frac{a_{20}}{1+b_{20}J_1} - b_{20}\omega_3 - \frac{\zeta_0 J_1}{1+b_{20}J_1} < 0$, and $\delta_1\theta P + \delta_2(1-\theta)P < 0$. Let $\delta_3 = \max\{\delta_1, \delta_2\}$ and $P_1 = (n_3+n_4)P$. Then there exists $t_2 \in [t^*, t^*+P_1]$ such that $x_0(t_2) \geqslant \omega_3$. If instead $x_0(t) < \omega_3$ for all $t \in [t^*, t^*+P_1]$, then Equation (6) holds. That is, for sufficiently large t, we have

$$x_1(t) \leqslant x_4(t) \leqslant \widetilde{x}_4(t) + \varepsilon.$$

Furthermore, Equation (7) holds on the interval $[t^*+n_3P, t^*+P_1]$. Therefore,

$$x_0((n_2+n_3+n_4+\theta)P^+) \geqslant \rho_2^{n_4}x_0((n_2+n_3+\theta)P^+).$$

From system (2), we obtain the following:

$$\begin{cases} \frac{\mathrm{d}x_0}{\mathrm{d}t} \geqslant x_0(t)\delta_3 & ,t \neq (n+\theta)P \\ \Delta x_0 \geqslant -\frac{h_{\max}}{\alpha_0}x_0 & ,t = (n+\theta)P \end{cases}. \tag{8}$$

On the interval $[t^*, t^*+n_3P]$, integrate the system (8), we have

$$x_0((n_2+n_3+\theta)P^+) \geqslant \left(1 - \frac{h_{\max}}{\alpha_0}\right)^{n_3} x_0((n_2+\theta)P^+)e^{n_3\delta_3 P}.$$

In conclusion, the following inequality can obtain

$$x_0((n_2+n_3+n_4+\theta)P^+) \geqslant \rho_2^{n_4}\left(1 - \frac{h_{\max}}{\alpha_0}\right)^{n_3} x_0((n_2+\theta)P^+)e^{n_3\delta_3 P} = \rho_3\omega_3 > \omega_3,$$

which is a contradiction. Hence, there exists $t_2 \in [t^*, t^* + P_1]$ such that $x_0(t_2) \geqslant \omega_3$. Let $t_1^* = \inf_{t \geqslant t^*}\{x_0(t) \geqslant \omega_3\}$. For any $t \in [t^*, t_1^*]$, we have $x(t) < \omega_3$, then, by integrating (8) on interval $[t^*, t_1^*)$, we gain $x_0(t) \geqslant \omega_3\left(1 - \frac{h_{\max}}{\alpha_0}\right)^{n_2+n_3} e^{(n_2+n_3)\delta_3 P} = \omega_1$. For $t \geqslant t_1^*$, the same discussion can be carried out. Therefore, $x_0(t) \geqslant \omega_1$ holds for $t \geqslant t_1$.

Case B: $t^* \neq (n_2 + \theta)P$. Assume $t^* \in ((n_2' + \theta)P, (n_2' + \theta + 1)P]$. The following analysis presents possible outcomes in this scenario.

Case B(a): For all $t \in (t^*, (n_2' + 1 + \theta)P)$, we have $x_0(t) < \omega_3$. Following the approach in Case A, there exists $t_2' \in [(n_2' + \theta + 1)P, (n_2' + n_3' + n_4' + 1 + \theta)P]$ such that $x_0(t_2') \geqslant \omega_3$. This proof is omitted here. Define $t_2'' = \inf_{t \geqslant t^*}\{x_0(t) \geqslant \omega_3\}$. For any $t \in (t^*, t_2'')$, we have $x_0(t) < \omega_3$. Integrating Equation (8) over (t^*, t_2'') gives $x_0(t) \geqslant x_0(t^*)\left(1 - \frac{h_{\max}}{\alpha_0}\right)^{n_3'+n_4'} e^{\delta_3(n_3'+n_4'+1)P} = \omega_1$.

Case B(b): There exists $t_0' \in (t^*, (n_2' + 1 + \theta)P)$ such that $x_0(t_0') \geqslant \omega_3$. Define $t_3' = \inf_{t \geqslant t^*}\{x_0(t) \geqslant \omega_3\}$. Then for all $t \in (t^*, t_3')$, we have $x_0(t) < \omega_3$. Integrating Equation (8) over the interval (t^*, t_3') yields $x_0(t) \geqslant x_0(t^*)e^{\delta_3 P} = \omega_1$. Since $x_0(t_3') \geqslant \omega_3$, the same argument applies for $t \geqslant t_3'$. Therefore, $x_0(t) \geqslant \omega_1$ for all $t \geqslant t_1$. This completes the proof. □

4. Numerical Simulation and Discussion

This section employs numerical simulations to achieve three progressive goals: the first is to visually validate the theoretical results of Section 3, demonstrating both extinction and permanence scenarios (Figures 1 and 2). In Figure 1, panels (a), (b), and (c) illustrate the prey population's decay to zero, the predator's convergence to a periodic oscillation, and the phase portrait converging to the boundary solution $(0, \widetilde{x}_1(t))$, respectively, thereby visually confirming Theorem 2. Conversely, Figure 2 shows sustained oscillations of both populations, confirming Theorem 2. The second aim is to identify the key management levers by quantifying the influence of major parameters on the extinction threshold condition R_0 through sensitivity analysis (Figures 3–5). Figure 3 (the PRCC plot) ranks parameter sensitivity, while Figures 4 and 5 illustrate two-parameter interactions, demonstrating how P, $h_{\max}$, and d_1 jointly modulate R_0. The third aim is to explore complex dynamics beyond the stable regimes, revealing potential risks (e.g., chaotic fluctuations) that may arise from suboptimal management, thereby extending the theoretical findings to more realistic contexts (Figures 6–13). Figures 6–13 systematically reveal bifurcations, period-doubling cascades, chaotic transitions, and the coexistence of attractors as key parameters (P, $h_{\max}$, α, d_1) are varied, providing a dynamical map of potential management outcomes.

First, to validate the global stability condition in Theorem 2, we select parameters satisfying (H_1), (H_2'), and (H_3). The following numerical simulations, conducted using MATLAB R2024a, validate the established theoretical results. The parameter values used in our model are $a_{10} = 0.6$, $b_{10} = 0.8$, $a_{11} = 3$, $b_{11} = 0.25$, $h_{\max} = 0.75$, $\alpha_0 = 0.76$, $d_0 = 0.11$, $a_{20} = 0.4$, $b_{20} = 0.2$, $\alpha = 15$, $\zeta_0 = 0.8$, $A_0 = 0.05$, $B_0 = 1$, $d = 0.25$, $c_0 = 0.85$, $d_1 = 0.11$, $P = 2$, $\theta = 0.25$. The calculations yield the following values:

$$H_1 = (1 - d_0)(1 - d_1)e^{a_{11}\theta P - d(1-\theta)P} = 2.4398 > 1,$$

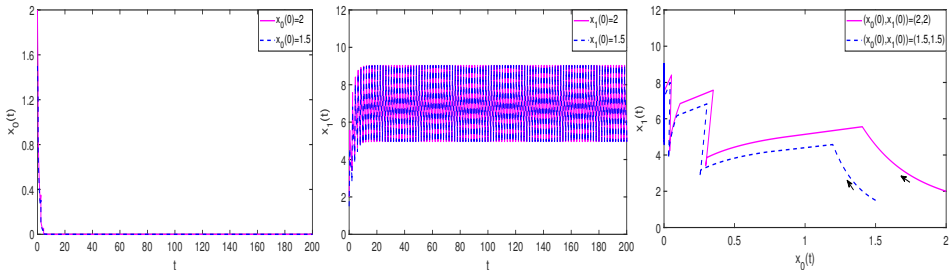
$$H_2' = (1 - d_0)(1 - d_1)e^{a_{11}\theta P - d(1-\theta)P - 2\ln\frac{a_{11} + b_{11}x^*(e^{a_{11}\theta P} - 1)}{a_{11}}} = 0.9338 < 1,$$

and

$$H_3 = \left(1 - \frac{h_{\max}}{\alpha_0 + b}\right)\exp(a_{10}\theta P + d(1-\theta)P + \mathbf{M}) = 0.2327 < 1.$$

These results satisfy the conditions of Theorem 2. We therefore conclude that the periodic solution $(0, \widetilde{x}_1(t))$ of system (2) is globally asymptotically stable (see Figure 1).

As shown in Figure 1, when the parameters satisfy the conditions of Theorem 2, the system exhibits typical prey-extinction dynamics. Specifically, Figure 1a shows that the prey population $x_0(t)$ rapidly decays to zero. Meanwhile, Figure 1b demonstrates that the predator population $x_1(t)$ tends toward a stable periodic oscillation; this behavior is manifested in the phase portrait in Figure 1c as the system trajectory converges to the boundary periodic solution $(0, \widetilde{x}_1(t))$. When the parameters satisfy the conditions of Theorem 2, the system trajectory converges to the boundary periodic solution $(0, \widetilde{x}_1(t))$, i.e., the prey population goes extinct while the predator population exhibits periodic fluctuations. This numerically verifies the theoretically derived extinction threshold. From a management perspective, this state simulates the ecological scenario where, after the collapse of the target fish resource, predation pressure—though at a low level—still recurs periodically due to the existence of migratory sources, thereby highlighting the extreme importance of maintaining the fish population above the extinction threshold.



(**a**) Time series of $x_0(t)$. (**b**) Time series of $x_1(t)$. (**c**) Phase portrait of $(0, \widetilde{x}_1(t))$.

Figure 1. This figure validates the global asymptotic stability of the prey-extinction periodic solution $(0, \widetilde{x}_1(t))$ as stated in Theorem 2. Panel (**a**) shows the time series of the prey population $x_0(t)$, which rapidly decays to zero, confirming prey extinction. Panel (**b**) displays the time series of the predator population $x_1(t)$, which converges to a stable periodic oscillation; this oscillation is driven by the predator's intrinsic growth during migration-free periods and mortality during migration, as described by subsystem (3). Panel (**c**) presents the phase portrait, where the system trajectory converges to the boundary periodic solution $(0, \widetilde{x}_1(t))$, visually confirming the stability of the prey-extinct state. The basic parameters are $a_{10} = 0.6$, $b_{10} = 0.8$, $a_{11} = 3$, $b_{11} = 0.25$, $h_{\max} = 0.75$, $\alpha_0 = 0.76$, $d_0 = 0.11$, $a_{20} = 0.4$, $b_{20} = 0.2$, $\alpha = 15$, $\zeta_0 = 0.8$, $A_0 = 0.05$, $B_0 = 1$, $d = 0.25$, $c_0 = 0.85$, $d_1 = 0.11$, $P = 2$, $\theta = 0.25$.

Next, to demonstrate the permanence condition in Theorem 3, a different parameter set is used. Setting the model parameters as $a_{10} = 0.8$, $b_{10} = 0.15$, $a_{11} = 0.9$, $b_{11} = 0.27$, $h_{\max} = 0.25$, $\alpha_0 = 0.9$, $d_0 = 0.13$, $a_{20} = 0.5$, $b_{20} = 0.22$, $\alpha = 0.9$, $\zeta_0 = 0.4$, $A_0 = 0.04$, $B_0 = 0.3$, $d = 0.2$, $c_0 = 0.1$, $d_1 = 0.27$, $P = 3$, $\theta = 0.35$, then

$$H_1 = (1 - d_0)(1 - d_1)e^{a_{11}\theta P - d(1-\theta)P} = 1.4284 > 1,$$

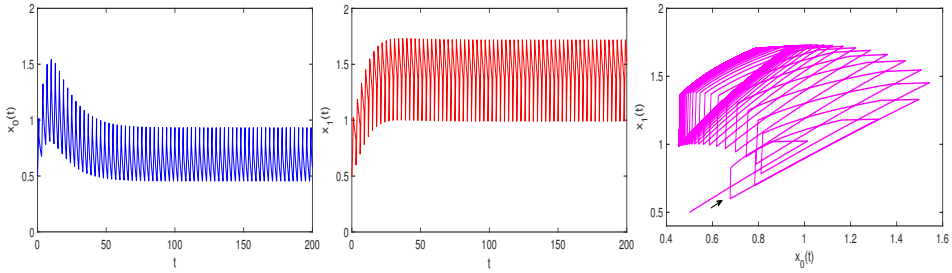
and

$$H_4 = \left(1 - \frac{h_{\max}}{\alpha_0}\right) \exp(a_{10}\theta P + d(1-\theta)P + \mathbf{M}') = 5.3298 > 1,$$

which implies that Theorem 3 is satisfied; therefore, system (2) is permanent (see Figure 2).

To systematically rank the impact of key parameters on the extinction threshold R_0, we perform a PRCC (Partial Rank Correlation Coefficient) analysis. This leads us to determine the critical factors governing the prey extinction threshold condition R_0, and

$$R_0 = \left(1 - \frac{h_{\max}}{\alpha_0}\right) \exp\left(a_{10}\theta P + d(1-\theta)P + \mathbf{M}'\right).$$



(**a**) Time series of $x_0(t)$. (**b**) Time series of $x_1(t)$. (**c**) Phase portrait.

Figure 2. This figure demonstrates the permanence of system (2) as established in Theorem 3. Panel (**a**) shows the time series of the prey population $x_0(t)$, which persists at positive levels with oscillations. Panel (**b**) shows the corresponding predator population $x_1(t)$ also persisting with oscillations. Panel (**c**) displays the phase portrait in which the trajectory remains bounded within a positive region of the state space, confirming the long-term coexistence of both populations. The basic parameters are $a_{10} = 0.8$, $b_{10} = 0.15$, $a_{11} = 0.9$, $b_{11} = 0.27$, $h_{\max} = 0.25$, $\alpha_0 = 0.9$, $d_0 = 0.13$, $a_{20} = 0.5$, $b_{20} = 0.22$, $\alpha = 0.9$, $\zeta_0 = 0.4$, $A_0 = 0.04$, $B_0 = 0.3$, $d = 0.2$, $c_0 = 0.1$, $d_1 = 0.27$, $P = 3$, and $\theta = 0.35$.

Predator fear levels toward prey α, predator migration mortality rate d_1, prey capture amount $h_{\max}$, pulse period P, and the inter-predator interference coefficient B_0 are selected as research parameters. To thoroughly explore the interactions among key parameters and their combined effects on the system's threshold condition R_0, we conduct a systematic sensitivity analysis of P, $h_{\max}$, d_1, and B_0 in this section. The PRCC values between each parameter and the prey extinction condition R_0 are calculated to evaluate their sensitivity. In this process, the LHS (Latin Hypercube Sampling) method is used to estimate the PRCC values, thereby accurately identifying the parameters that play a decisive role in the prey extinction threshold condition R_0 [44].

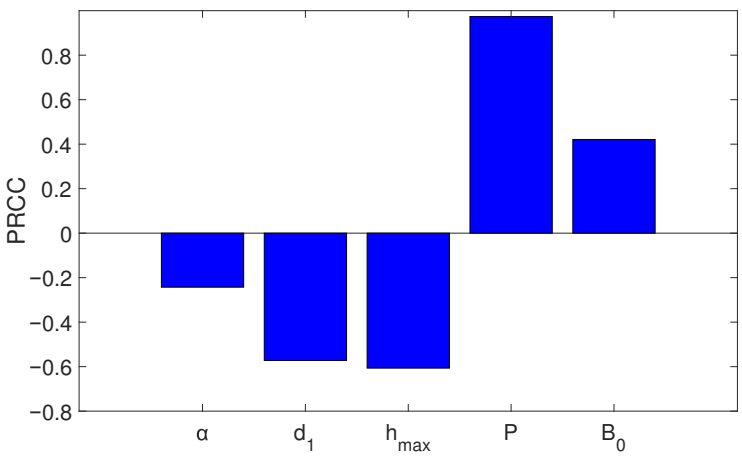
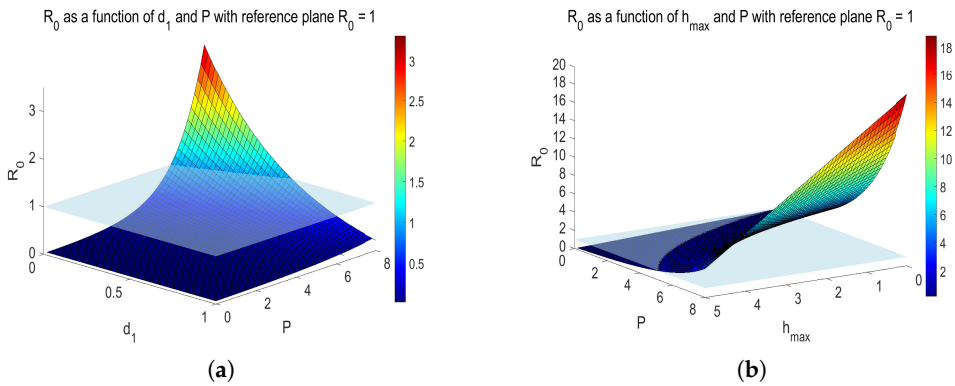


Figure 3. The PRCCs of the key parameters (α, d_1, $h_{\max}$, P, B_0) for the prey extinction threshold condition R_0. The basic parameters are $a_{10} = 5$, $b_{10} = 0.5$, $a_{11} = 2$, $b_{11} = 0.5$, $h_{\max} = 2$, $\alpha_0 = 6$, $d_0 = 0.2$, $a_{20} = 4$, $b_{20} = 0.2$, $\alpha = 0.3$, $\zeta_0 = 0.1$, $A_0 = 0.5$, $B_0 = 0.5$, $d = 0.2$, $c_0 = 0.1$, $d_1 = 0.2$, $P = 3$, and $\theta = 0.1$.

As observed in Figure 3, the pulse period P and the predator interference coefficient B_0 are positively correlated with the prey extinction threshold condition R_0. It has been demonstrated that an increase in P or B_0 raises R_0, which in turn contributes to an increase in prey density. Conversely, prey fear level α, predator migration mortality d_1, and maximum prey harvest $h_{\max}$ show negative correlations with R_0. An increase in these parameters decreases R_0, which ultimately reduces prey density. Further PRCC results indicate that the parameters d_1, $h_{\max}$, P, and B_0 all demonstrate substantial correlation

strengths, with absolute values exceeding 0.4. This demonstrates that these four parameters significantly influence R_0, thus serving as key regulators of prey population density. From a biological perspective, elevated predator migration mortality d_1 initially leads to a temporary decline in predator abundance. This reduction in density-dependent stress subsequently increases reproductive success and juvenile survival, ultimately triggering a compensatory population surge in the following year. The reinvasion of these recovered predators into prey habitats consequently drives prey population suppression through intensified predation pressure.



(**a**) (**b**)

Figure 4. The importance of predator migration mortality rate d_1, prey capture amount $h_{\max}$, and impulsive period P on the threshold condition R_0. (**a**) Effect of predator migration mortality rate d_1 and impulsive period P on R_0 with $h_{\max} = 5.9$. (**b**) Effect of prey capture amount $h_{\max}$ and impulsive period P on R_0 with $d_1 = 0.9$.

As shown in Figure 4, the impulsive period P, maximum prey harvest amount $h_{\max}$, and predator migration mortality rate d_1 strongly influence the threshold condition R_0. A higher impulsive period P leads to a higher R_0. A higher predator migration mortality rate d_1 also increases R_0. In contrast, a lower maximum prey harvest amount $h_{\max}$ decreases R_0. Therefore, harvesting prey less frequently helps to maintain a larger prey population. The two-parameter analysis (Figure 4) demonstrates that the positive effect of the pulse period P on R_0 is more pronounced at low d_1 or high $h_{\max}$. This finding suggests that tuning the frequency of management interventions (P) is of particular importance in contexts characterized by low predator migration mortality or intense fishing pressure.

Figure 5 reveals that increasing the predator interference coefficient B_0 significantly mitigates top-down control on prey populations. The magnitude of this effect is strongly modulated by predator migration mortality d_1, with minimal mortality leading to maximal prey survival benefits. Ecological interpretation suggests that low d_1 maintains robust predator populations wherein interference competition B_0 becomes particularly effective at reducing per-capita predation rates. This parameter interaction creates a novel ecological scenario where prey thrive most when predator populations experience both low migration mortality and strong intra-specific interference, highlighting non-linear dependencies in food web regulation. The one-parameter analysis (Figure 5) and its interaction trend with d_1 demonstrate that the regulatory role of predator intraspecific interference B_0 in population dynamics is highly dependent on the intrinsic pressure of the predator population (as indicated by d_1), uncovering the nonlinear nature of top-down control processes. The sensitivity and two-parameter analyses collectively demonstrate that the effects of key management levers (e.g., P, $h_{\max}$, d_1) are rarely independent. Their interactions can amplify or mitigate each other's impact on the extinction threshold R_0, underscoring the necessity for integrated, multi-parameter management strategies rather than isolated adjustments of a single factor.

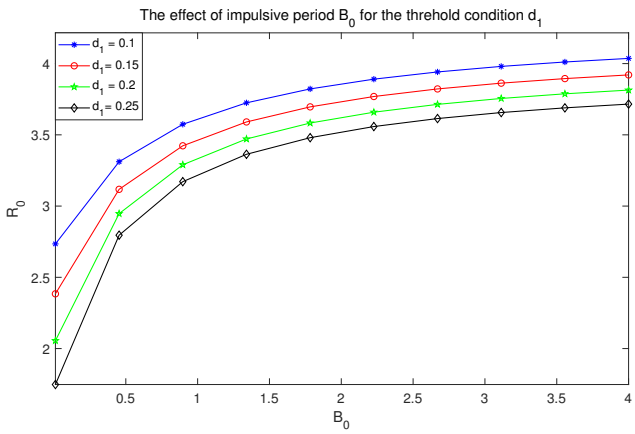
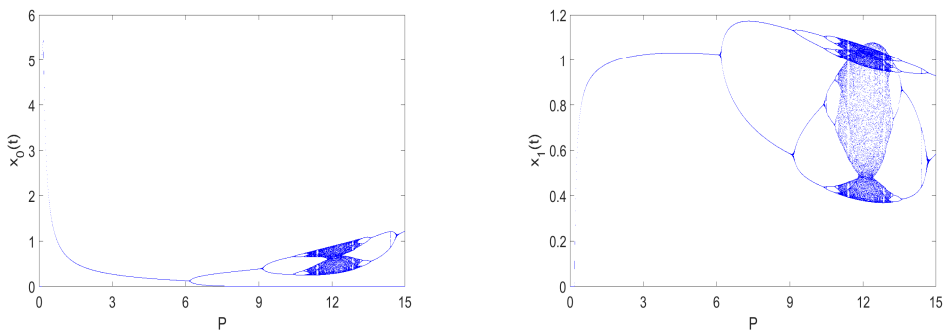


Figure 5. Parameter dependence of the threshold condition R_0. Setting the model parameters as $a_{10} = 5$, $b_{10} = 0.5$, $a_{11} = 2$, $b_{11} = 0.5$, $h_{\max} = 2$, $\alpha_0 = 6$, $d_0 = 0.2$, $a_{20} = 4$, $b_{20} = 0.2$, $\alpha = 0.3$, $\zeta_0 = 0.1$, $A_0 = 0.5$, $d = 0.2$, $c_0 = 0.1$, $P = 3$, $\theta = 0.1$.

Finally, we explore the complex dynamical behaviors that can emerge outside the stability regimes, taking the pulse period P as a representative example. We proceed to analyze several significant parameters governing the behavioral dynamics of system (2). The following analysis focuses on the role of the impulse period P in governing the dynamics of system (2). Setting the parameters as $a_{10} = 0.5$, $b_{10} = 0.1$, $a_{11} = 0.3$, $b_{11} = 0.3$, $h_{\max} = 0.2$, $\alpha_0 = 0.5$, $d_0 = 0.15$, $a_{20} = 3$, $b_{20} = 0.5$, $\alpha = 0.1$, $\zeta_0 = 3.5$, $A_0 = 0.2$, $B_0 = 0.3$, $d = 0.1$, $c_0 = 0.3$, $d_1 = 0.15$, and $\theta = 0.3$, we can derive the bifurcation diagrams of predator–prey model (2) with a bifurcation parameter P from 0.01 to 15.



(**a**) (**b**)

Figure 6. Bifurcation diagrams of system (2) with $0 \leqslant P \leqslant 15$. (**a**) Bifurcation diagram of $x_0(t)$. (**b**) Bifurcation diagram of $x_1(t)$. Setting the parameters as $a_{10} = 0.5$, $b_{10} = 0.1$, $a_{11} = 0.3$, $b_{11} = 0.3$, $h_{\max} = 0.2$, $\alpha_0 = 0.5$, $d_0 = 0.15$, $a_{20} = 3$, $b_{20} = 0.5$, $\alpha = 0.1$, $\zeta_0 = 3.5$, $A_0 = 0.2$, $B_0 = 0.3$, $d = 0.1$, $c_0 = 0.3$, $d_1 = 0.15$, and $\theta = 0.3$.

Figure 6 illustrates the complex bifurcation scenario of system (2) under varying impulsive periods P. An increase in P triggers a classical route to chaos through successive period-doubling bifurcations, followed by a reverse sequence of period-halving (inverse period-doubling) bifurcations that restore system order. This complete bifurcation cascade demonstrates the fundamental role of the impulsive period in regulating system dynamics between order and chaos. Notably, chaos emerges in the system when P increases to approximately 12.8 (Figure 8a); as P further rises to 15, the system reverts to a stable P-periodic solution via inverse period-doubling bifurcation. This indicates that there exists a 'chaotic window' for the interval of management pulses (P), where an inappropriate interval may lead to unpredictable drastic fluctuations in population size. Supplementary Figures 7 and 8 provide concrete examples of the characteristic behaviors observed during these transitions, numerically verifying the theoretical predictions.

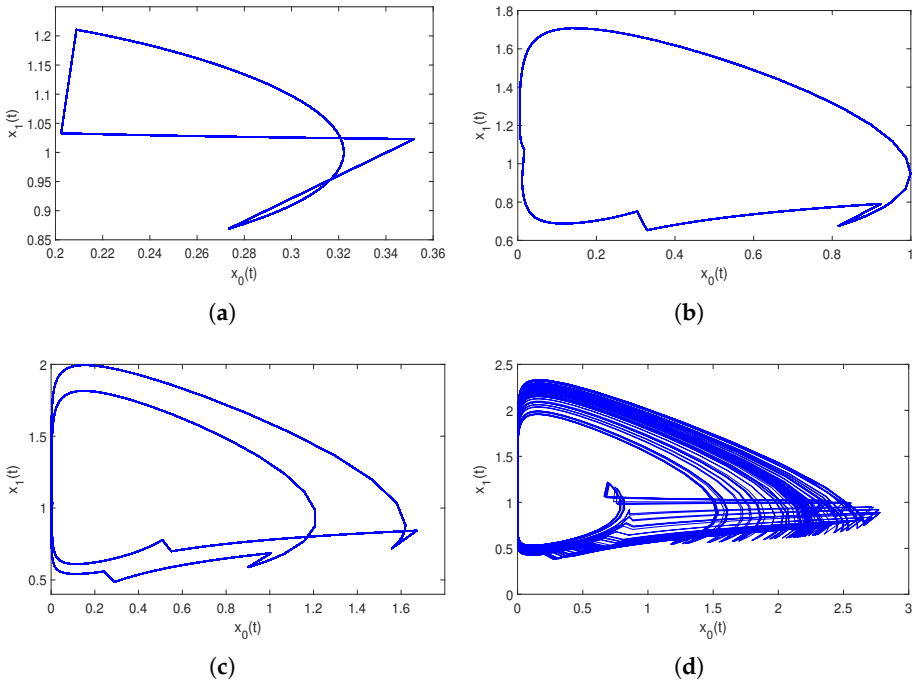


Figure 7. A period-doubling cascade transitions the system from a P-periodic solution to a $4P$-periodic solution. (**a**) Phase portrait of P-periodic solution when $P = 4$. (**b**) Phase portrait of $2P$-periodic solution when $P = 8$. (**c**) Phase portrait of $4P$-periodic solution when $P = 9.6$. (**d**) Phase portrait of chaos when $P = 12$.

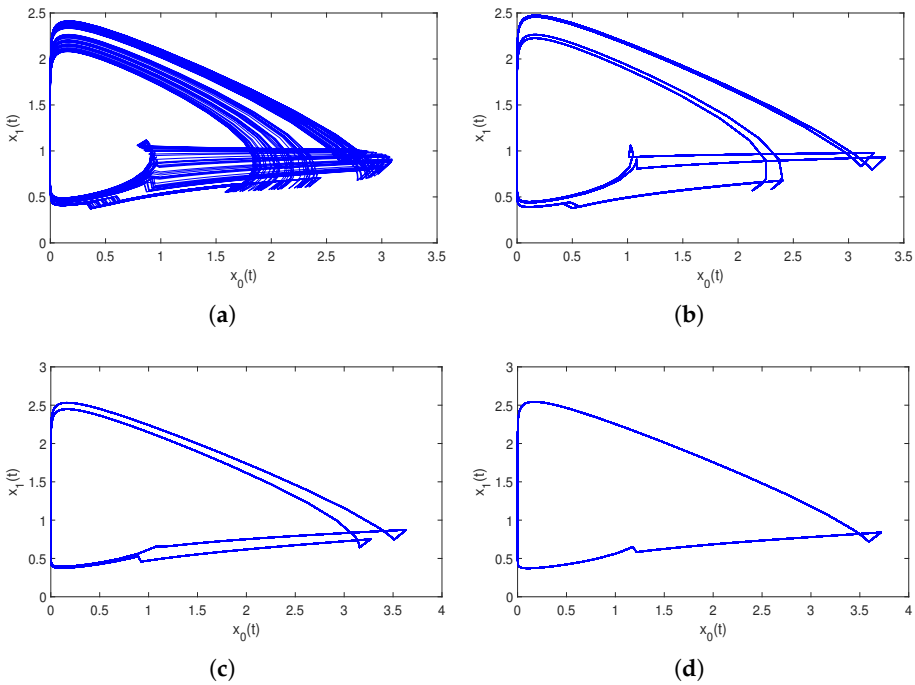
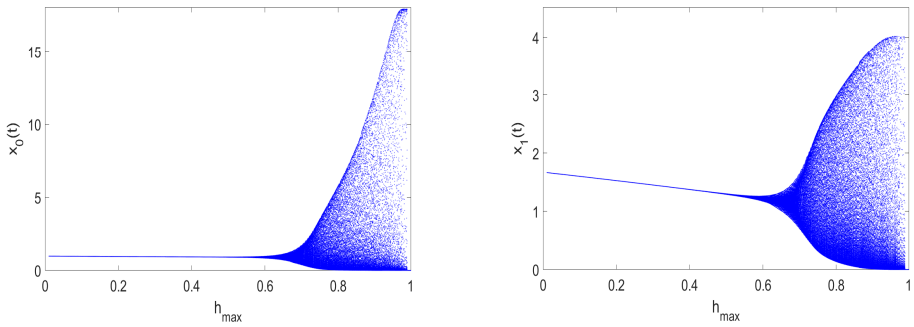


Figure 8. Period-halving bifurcation reduces the system to a P-periodic solution. (**a**) Phase portrait exhibiting chaotic behavior when $P = 12.8$. (**b**) Phase portrait of the $4P$-periodic solution when $P = 13.5$. (**c**) Phase portrait of the $2P$-periodic solution when $P = 14.5$. (**d**) Phase portrait of the P-periodic solution when $P = 15$.

Next, we investigate the influence harvesting ability $h_{\max}$ of the prey for system (2). Set the parameters of system (2) as $a_{10} = 0.4$, $b_{10} = 0.1$, $a_{11} = 0.25$, $b_{11} = 0.25$, $\alpha_0 = 0.9$, $d_0 = 0.1$, $a_{20} = 2.8$, $b_{20} = 0.15$, $\alpha = 0.2$, $\zeta_0 = 2.5$, $A_0 = 0.2$, $B_0 = 1$, $d = 0.1$, $c_0 = 0.32$,

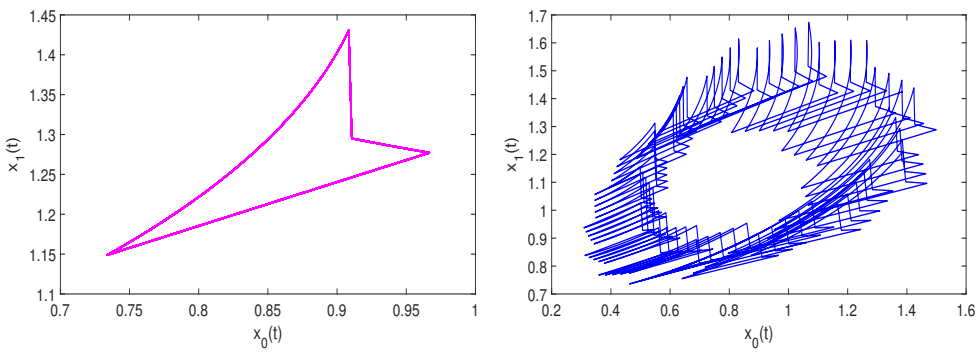
$d_1 = 0.1$, $P = 1$, $\theta = 0.2$, with $x_0(0) = 0.6$, $x_1(0) = 0.4$. Figure 9 shows the bifurcation diagrams of system (2) based on the bifurcation parameter $h_{\max} > 0$.



(**a**) (**b**)

Figure 9. Bifurcation diagrams of system (2) with the maximum harvest amount $h_{\max}$ as the bifurcation parameter (for $0.01 \leqslant h_{\max} < 1$). (**a**) Bifurcation diagram of $x_0(t)$ for bifurcation parameter $0.01 \leqslant h_{\max}$. (**b**) Bifurcation diagram of $x_1(t)$ for bifurcation parameter $0.01 \leqslant h_{\max}$. Set the parameters of system (2) as $a_{10} = 0.4$, $b_{10} = 0.1$, $a_{11} = 0.25$, $b_{11} = 0.25$, $\alpha_0 = 0.9$, $d_0 = 0.1$, $a_{20} = 2.8$, $b_{20} = 0.15$, $\alpha = 0.2$, $\zeta_0 = 2.5$, $A_0 = 0.2$, $B_0 = 1$, $d = 0.1$, $c_0 = 0.32$, $d_1 = 0.1$, $P = 1$, and $\theta = 0.2$.

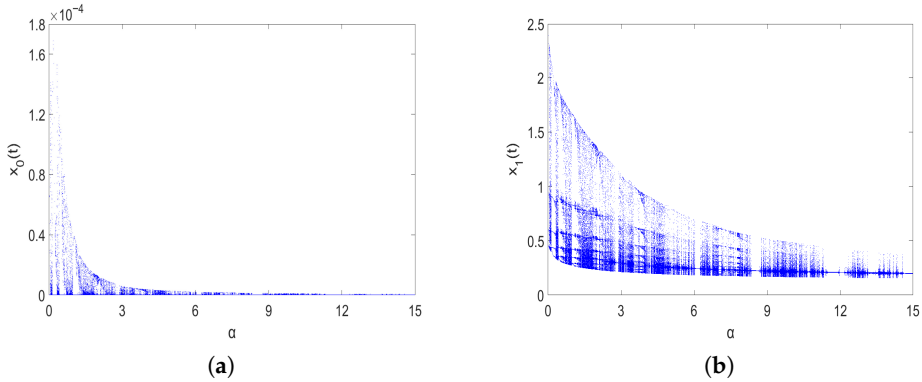
As shown in the bifurcation diagram, the maximum amplitude of system dynamics remains relatively stable. Figure 10 shows that with the increase in fishing intensity $h_{\max}$, the system undergoes an abrupt transition from a P-periodic solution to chaotic behavior, which represents the occurrence of a crisis-induced chaotic transition. Specifically, when $h_{\max}$ exceeds the critical value ($\sim$0.6), the system completely loses stability, ultimately leading to the extinction of both prey and predator populations (i.e., population collapse). This result provides a key dynamical basis for setting the "maximum allowable catch" in fishery management.



(**a**) (**b**)

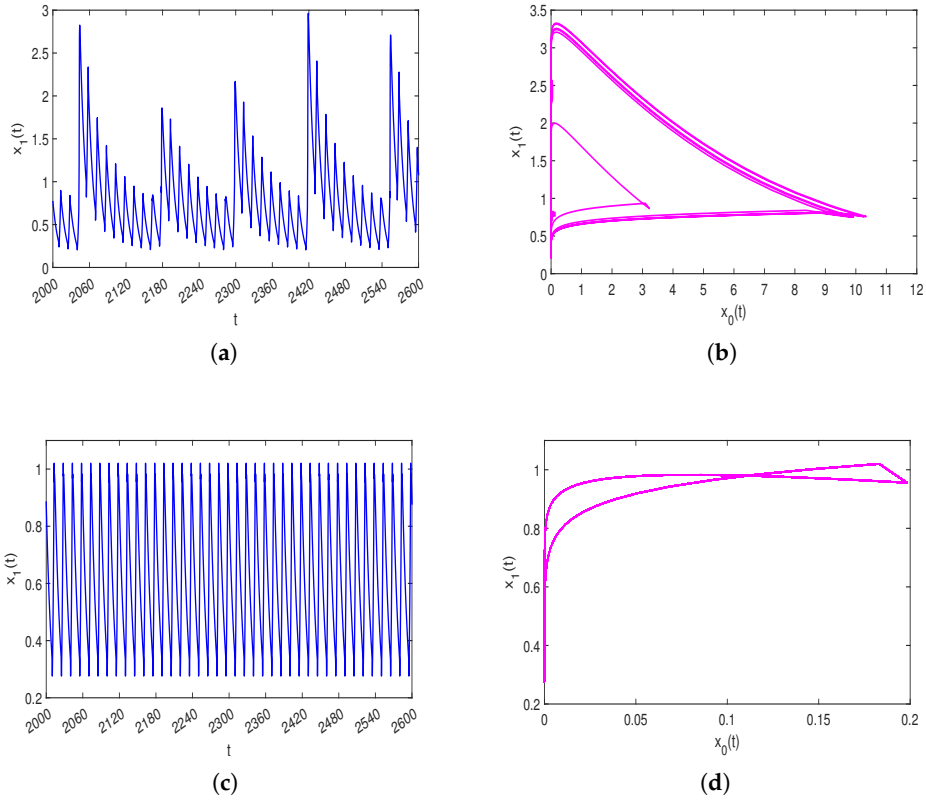
Figure 10. System (2) suddenly changes from a P-periodic solution to chaos. (**a**) Phase portrait of P-periodic solution when $h_{\max} = 0.5$. (**b**) Phase portrait of chaos when $h_{\max} = 0.7$.

Suppose the parameters of system (2) are $a_{10} = 5$, $b_{10} = 0.1$, $a_{11} = 0.5$, $b_{11} = 0.1$, $h_{\max} = 0.1$, $\alpha_0 = 0.5$, $d_0 = 0.08$, $a_{20} = 2$, $b_{20} = 0.5$, $\zeta_0 = 5$, $A_0 = 0.01$, $B_0 = 1$, $d = 0.1$, $c_0 = 0.5$, $d_1 = 0.15$, $P = 15$, $\theta = 0.2$. Bifurcation diagrams of system (2) are computed for parameter α in the range [0.01, 15] (see Figure 11). Figure 11 shows that different attractors will coexist for the same fear level alpha. Figure 12 demonstrates that strange attractors coexist with a $4P$-periodic solution at $\alpha = 3.72$.



(**a**) (**b**)

Figure 11. Bifurcation diagram of system (2) with fear level α as the bifurcation parameter. (**a**) Bifurcation diagram of $x_0(t)$. (**b**) Bifurcation diagram of $x_1(t)$. Suppose the parameters of system (2) are $a_{10} = 5$, $b_{10} = 0.1$, $a_{11} = 0.5$, $b_{11} = 0.1$, $h_{\max} = 0.1$, $\alpha_0 = 0.5$, $d_0 = 0.08$, $a_{20} = 2$, $b_{20} = 0.5$, $\zeta_0 = 5$, $A_0 = 0.01$, $B_0 = 1$, $d = 0.1$, $c_0 = 0.5$, $d_1 = 0.15$, $P = 15$, and $\theta = 0.2$.

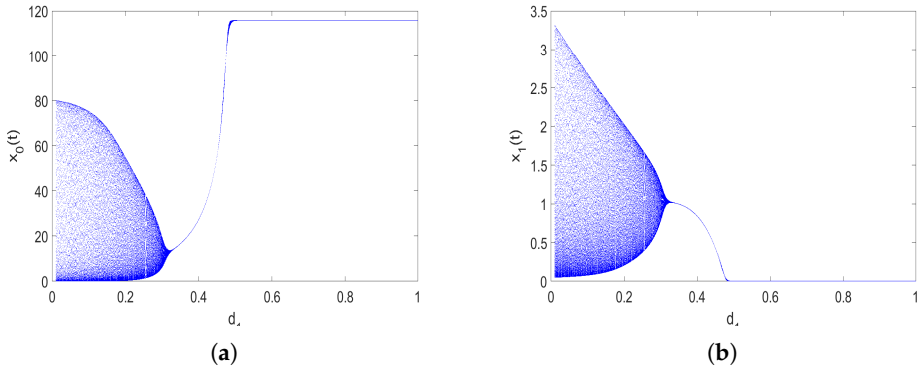


(**a**) (**b**)

(**c**) (**d**)

Figure 12. When $\alpha = 3.72$, the $4P$-periodic solution and P-periodic solution coexist. (**a**) Time series of $x_1(t)$ with $(x_0(0), x_1(0) = (0.6, 0.4))$. (**b**) The solution with $(x_0(0), x_1(0) = (0.6, 0.4))$ tends to a $4P$-periodic solution. (**c**) Time series of $x_1(t)$ with $(x_0(0), x_1(0) = (2, 2))$. (**d**) The solution with $x_0(0) = 2, x_1(0) = 2$ tends to a $4P$-periodic solution.

Similarly to Figures 6, 9 and 11, we set the model parameters as $a_{10} = 1$, $b_{10} = 0.2$, $a_{11} = 0.1$, $b_{11} = 0.1$, $h_{\max} = 0.2$, $\alpha_0 = 0.5$, $d_0 = 0.1$, $a_{20} = 2$, $b_{20} = 0.01$, $\alpha = 1$, $\zeta_0 = 2$, $A_0 = 0.2$, $B_0 = 0.3$, $d = 0.1$, $c_0 = 0.1$, $P = 1$, and $\theta = 0.1$. We obtained the bifurcation diagram of system (2) with respect to the bifurcation parameter d_1 ranging from 0.01 to 1 (see Figure 13). Figure 13 shows that as the predator migration mortality rate d1 increases, the system will transition from chaos to a P-periodic solution.



(**a**) (**b**)

Figure 13. Bifurcation diagrams of system (2) with the predator migration mortality rate d_1 as the bifurcation parameter (for $0.01 \leqslant d_1 \leqslant 1$). (**a**) Bifurcation diagram of $x_0(t)$ for the bifurcation parameter d_1. (**b**) Bifurcation diagram of $x_1(t)$ for the bifurcation parameter d_1. Set the model parameters as $a_{10} = 1$, $b_{10} = 0.2$, $a_{11} = 0.1$, $b_{11} = 0.1$, $h_{\max} = 0.2$, $\alpha_0 = 0.5$, $d_0 = 0.1$, $a_{20} = 2$, $b_{20} = 0.01$, $\alpha = 1$, $\zeta_0 = 2$, $A_0 = 0.2$, $B_0 = 0.3$, $d = 0.1$, $c_0 = 0.1$, $P = 1$, and $\theta = 0.1$.

When the mortality rate of predators, denoted by d_1, is minimal, predators exert a substantial influence on their prey. The population dynamics of both species demonstrate chaotic behaviour. When the value of d_1 exceeds the critical value of approximately 0.47, a significant proportion of predators perish. The growth rate of prey populations increases exponentially before stabilising, a phenomenon that can be attributed to a significant decrease in predation pressure. A decline in population numbers or even extinction is often the consequence of excessively high mortality rates.

From a management perspective, the discovery of complex dynamical behaviors (e.g., bifurcations and chaos) carries significant implications. It indicates that within specific parameter regions, the long-term evolutionary behavior of the system is highly sensitive to initial conditions and parameter perturbations, which significantly increases the difficulty and potential risks of practical management. Therefore, in the practice of fishery resource management, efforts should be made to avoid setting system parameters within these "sensitive regions" to reduce the unpredictability of population dynamics.

In summary, the exploration beyond stable regimes reveals that the system can exhibit rich dynamics, including periodic doubling, chaos, and the coexistence of attractors, particularly when key parameters like the pulse period P, harvesting intensity $h_{\max}$, or fear level α exceed certain ranges. These findings provide crucial additional insights: (1) They delineate "danger zones" in parameter space where management outcomes become highly unpredictable or sensitive to initial conditions. (2) The presence of chaotic or high-periodic oscillations implies that even in the absence of extinction, populations may undergo severe fluctuations that threaten economic stability and ecosystem health. (3) Therefore, effective management must consider not only the extinction threshold but also the broader dynamical landscape to avoid inadvertently pushing the system into a volatile, though persistent, state.

5. Conclusions

This study formulates a predator–prey model with a Beddington–DeAngelis functional response that incorporates prey fear, large-scale predator migration, and impulsive density-dependent nonlinear harvesting. Through theoretical analysis and numerical simulations, it provides theoretical support for sustainable resource management. The main contributions of this work are threefold: first, we theoretically derive explicit extinction and permanence thresholds (Theorems 1–3), which provide mathematically rigorous boundaries for sustainable harvesting. Second, through systematic numerical exploration (Figures 1–13), we

identify and quantify the key parameters (P, $h_{\max}$, d_1, B_0, α) that govern system dynamics, highlighting their interactions and nonlinear effects. Third, we uncover complex dynamical regimes (bifurcations, chaos) that emerge outside stability boundaries, revealing previously unrecognized risks in fishery management. These findings collectively advance the field of impulsive bioeconomic modeling by integrating multiple ecological mechanisms (fear, migration, nonlinear harvesting) and by providing both theoretical benchmarks and practical insights for adaptive management. Specifically, the results suggest that management strategies should simultaneously regulate harvesting intensity ($h_{\max}$), timing (P), and consider predator migration mortality (d_1) to maintain populations within safe dynamical regimes, thereby preventing both extinction and economically destabilizing fluctuations. The results demonstrate that the scientific and coordinated regulation of the maximum harvesting intensity $h_{\max}$ and the management intervention frequency pulse period P is the key pathway to steering the system towards permanence and avoiding the risk of prey extinction. Here, $h_{\max}$ establishes the foundation for an adaptive harvesting strategy based on population density, while P serves as a crucial temporal lever for balancing short-term economic benefits with long-term ecological stability. Future fishery management practices should strive to integrate this theoretical framework with field monitoring data to develop more precise and sustainable harvesting strategies.

We first prove the uniform boundedness of all solutions. Subsequently, we establish the local and global asymptotic stability conditions for the prey-extinction periodic solution $(0, \widetilde{x}_1(t))$, derive the criteria for system permanence, and determine the precise threshold condition for prey extinction. Second, numerical simulations were conducted. These not only confirmed the validity of the theoretical results but also identified the core influencing factors. Specifically, sensitivity analysis revealed four key parameters. The prey extinction threshold condition is primarily governed by four critical parameters: the predator migration mortality rate d_1, the maximum prey capture amount $h_{\max}$, the pulse period P, and the inter-predator interference coefficient B_0. A two-parameter sensitivity analysis demonstrated that both the pulse period P and interference coefficient B_0 are positively correlated with this threshold. Consequently, increasing either parameter strengthens the threshold condition, thereby enhancing the likelihood of prey population persistence. The predator migration mortality rate d_1 and maximum prey capture amount $h_{\max}$ showed a negative correlation with the prey extinction threshold R_0. Increases in these parameters pushed the prey population closer to extinction. These results suggest that extending the pulse period P and strengthening predator interference B_0 can improve prey survival chances. In contrast, high predator migration mortality d_1 or increased prey capture $h_{\max}$ will markedly elevate extinction risk. Therefore, management strategies should maintain appropriate pulse intervals and mitigate predator interference while avoiding excessive migration mortality and overharvesting. Finally, this study analyzed four key parameters including fear level α, predator migration mortality rate d_1, maximum prey capture amount $h_{\max}$, and pulse period P. These parameters significantly influenced the system state, with the pulse period P having a particularly strong effect. A suitable value of P could stabilize the population dynamics of both prey and predators. In addition, through systematic parameter interaction analysis, we found that the effect of the pulse period P is closely correlated with predator mortality d_1 and fishing intensity $h_{\max}$, which underscores the necessity of integrated management strategies. Exploration of the dynamics outside the stability domain revealed the existence of bifurcation and chaotic behaviors in the system, alerting us to the risk that inappropriate management may trigger unpredictable drastic fluctuations in populations. In summary, these results provide reliable theoretical support for fishery management. They offer guidance for scientifically regulating fishing effort,

protecting biodiversity, preventing overfishing, and ensuring the sustainable supply of fishery resources.

It should be emphasized that all of the above conclusions are derived under several simplifying assumptions, including deterministic impulsive harvesting, a fixed predator migration mortality rate, and a time-invariant representation of the prey fear response. These assumptions make it possible to obtain explicit extinction thresholds, stability domains, and bifurcation structures, but they inevitably restrict the range of dynamical behaviours that the model can capture. In real ecosystems, stochastic fluctuations in harvesting effort, migration losses, or behavioural responses may strongly interact with nonlinear functional responses and impulsive effects, thereby shifting extinction thresholds, modifying system robustness, and generating richer short-term dynamics such as noise-induced oscillations, transient population bursts, or regime shifts [24,45,46]. Consequently, the stability, permanence and bifurcation results obtained here should be interpreted as baseline predictions for systems in which environmental and management variability are moderate, and caution is required when extrapolating these findings to strongly stochastic settings.

Building on these limitations, many meaningful questions remain for future research. For example, how can we determine the optimal pulse control period P to achieve an appropriate balance between harvesting yield and ecological risk? If we consider stage structure in predators or prey, what kind of dynamic behavior will the system exhibit? If stochastic variability is introduced into capture intensity, migration loss, or fear responses, and additional mechanisms such as adaptive fear responses or cooperative hunting are incorporated, what kind of dynamic behavior will the system exhibit? Meanwhile, the current model primarily considers the impact of the fear effect on the prey population. In reality, the fear effect may also significantly influence predator behavior and functional response parameters. Investigating how fear shapes predator foraging strategies remains a meaningful direction for future research.

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